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Cow Face Recognition with Noise Labels Using Vision Transformer

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***Abstract*—Animal face recognition is a crucial step for people to monitor and manage livestock more accurately and conveniently, playing an important role in intelligent breeding. Traditionally, the method of labeling animals with a different type of marks such as tags, tattoos and brands is generally applied to uniquely identify individual livestock. However, the traditional way of identification is high cost, low efficiency and unreliable. In this paper, we propose a deep learning approach based on a framework of semi-supervised noise-label learning to realize noisy cow face recognition, which adopts the Vision Transformer model that has recently become popular in the filed of computer vision. Our network is trained on a dataset that consists of 6 individual cows with a total of 3556 facial images. Due to the lack of adequate datasets, this is a challenging task for achieving high performance. To evaluate the robustness of our approach to noise, we compared it with several other noisy learning methods. Our results demonstrate that Vision Transformer can be used effectively for cow face recognition, and can achieve good performance even in the presence of noise.**

***Index Terms*—Cow face recognition, Vision Transformer, Semi-supervised learning, Learning with noise labels**

I. Introduction

The identification of individual animals on the farm is a critical component of livestock management and has become an indispensable technology for animal husbandry in modern agriculture. To enhance yield and quality, farmers must be able to track and monitor individual animal performance and health in a more efficient way. Accurately identifying individual animals also allows farmers to comply with regulations and traceability requirements. Traditionally, several identification approaches have been developed for animal identification, including tattoos, ear tags and notch ears [1]. However, these methods have several limitations. One of the primary challenges is the risk of tag or mark loss. Physical tags may detach or be lost over time, while marks can fade or become obscured by dirt or hair. As a result, farmers have to re-tag and re-mark animals. Furthermore, there are usually thousands of animals need to be dealt with in real scenarios, which requires a substantial amount of time and resources, and may not guarantee effective identification.

To address the conundrum faced by traditional methods, electronic identification methods such as Radio Frequency Identification (RFID) tags have been widely used in livestock identification (Yang et al., 2020) [2]. However, these methods can be costly and may have adverse effects on the health of livestock. With the development of computer vision and machine learning, deep learning algorithms have emerged as a promising technology to address the issue of individual animal identification [3]. Among these technologies, face recognition holds great promise. One such method is the weighted Local Ternary Pattern (LTP), which is usually used for texture classification and has been shown to effectively capture facial features [4]. Another method that has been developed for face recognition is the String Grammar Nearest Neighbor (sgNN) algorithm [5]. Additionally, artificial neural network (ANN) [6] and radial basis function neural network (RBFNN) [7] have also utilized in face image recognition. While face recognition has been extensively used in human identification for many years [8], it has recently been adapted to identify animals. This technology can utilize the unique features of the face, such as the muzzle, eyes and ears, to achieve automatic individual cow identification from images or videos [9], streamlining the identification process and providing more accurate and reliable data. However, face recognition often requires a large amount of accurately labeled data to ensure accuracy and reliability, which can be difficult to obtain in real-world scenarios due to factors such as poor image quality, different annotation sources and class imbalance. In animal face recognition, label noise can significant impact the performance of the methods.

In this paper, we aim to explore the potential of deep learning methods for the challenging task of noise cow face recognition. In contrast to prior works, we adopt the ViT [10] model, which has recently become popular in the filed of computer vision, as our backbone network structure for cow face recognition instead of CNNs [11]. To tackle the issue of noise in the data, we propose to use semi-supervised noise-label learning methods to achieve robustness to the noise labels. Our experimental results show that the proposed approach based on this framework can achieve good performance even in the presence of noise. The key contributions of our work are as follow:

- We propose a noval approach to cow face recognition using the ViT model and make some improvements to enhance its performance.
- We adopt a noise recognition framework that converts noise learning into semi-supervised learning for cow face recognition.
- We implement a noise cow face recognition method and achieve good experimental results.

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II. Related work

A. Animal Face Recognition

In recent years, numerous studies have investigated deep learning approaches for animal face recognition especially livestock, but most existing methods are based on CNNs. For example, Hanen et al (2018) [12] developed a CNN model for pig face recognition from 1553 face images of 10 individual pigs, obtaining an accuracy of 96.7%. Similarly, Billah et al. (2022) [13] proposed a goat face recognition approach using a deep CNN, achieving an accuracy of 96.4% on a dataset of 3278 pictures with 10 individual goats.

Efforts have also been made to devise more efficient deep learning methods for animal face recognition. For instance, Xu et al. (2021) [14] presented a face identification framework based on the light-wight Retina-Mobilenet for cattle face recognition, which achieved remarkable recognition performance with an average precision of 91.3% and an average processing time of 0.042 seconds per image.

In contrast to the methods mentioned above, our approach is based on the ViT model and utilizes a series of 3×3 convolution modules to generate patch embeddings that possess more image-specific inductive bias. To mitigate the overfit of ViT on small dataset, we have adjusted the hyper parameters of our ViT model accordingly. Moreover, we train our ViT model from scratch.

B. Learning with Noisy Labels

As mentioned in Subsect.II-A, the majority of methods for animal face recognition rely on clean labels. However, obtaining a large amount of image data without noise is often challenging in practice, whether through manual or machine annotation methods. Thus, addressing the issue of noisy animal face recognition is necessary. For learning with noisy labels, various approaches have been proposed. Han et al. (2018) [15] introduced a new paradigm, Co-teaching, that trains two deep neural networks to teach each other. Ma et al. (2020) [16] proposed a new framework to build robust loss functions called APL, which combines two normalized loss functions. Li et al. (2020) [17] proposed DivideMix, a method that leverages semi-supervised learning to address noisy labels by dynamically dividing the noisy data into labeled and unlabeled samples. Wang et al. (2020) [18] concluded a versatile framework, noisy labels learning by semi-supervised learning (SemiNLL), which combines sample selection (SS) and semi-supervised learning (SSL) in an end-to-end scheme.

Overall, the study of noisy animal face recognition is a relatively new and rapidly developing field. Our method seeks to use the existing noisy labels learning methods to address the issue of noisy labels and improve the performance of animal face recognition in the presence of noise.

III. Methods

A. Network Architecture

Our Transformer model follows the architecture of ViT, which employs the encoder component of the original Transformer model. The key deviation between our model and the original ViT model is the approach we used to generate patch embeddings. Instead of directly extracting them from the input image, we apply a series of 3×3 convolution modules followed by batch normalization and max-pooling, to obtain inductive bias, as depicted in Fig.1. Specifically, our model takes images of cow faces with the resolution of (H, W, C) as input. The input passes through two convolution modules with a stride of 2 followed by a batch normalization layer and a max-pooling layer. After that, it undergoes three convolution module with a stride of 1 followed by a batch normalization layer, and is eventually resized into a sequence of patch embeddings. The first 4 convolution modules are activated by the GELU function after batch normalization. After the first convolution module, the number of image channels C = 3 increases to dim/16, and for the subsequent 4 convolution modules, the number of channels increases to dim, where dim represents the dimension of each patch.

As ViT did, we prepend a learnable embedding to the sequence of embedded patches $\left(z_0^0 = x_{class}\right)$, whose state at the output of the Transformer encoder (z_L^0) is the final cow face image embedding. Additionally, we still add position embeddings to the patch embeddings to retain positional information despite their potential redundancy due to the use of convolution module to generate patch embeddings, as shown in Equation (1). The patch embeddings then serve as input to the Transformer Encoder, which consists of multi-headed self-attention (MSA) and MLP blocks. LayerNorm (LN) is applied before every block and residual connections are applied after every block, as illustrated in Fig.1.

$$z_0 = \left[x_{class}; x_p^1; x_p^2; \cdots; x_p^N\right] + E_{pos}, \quad E_{pos} \in R^{(N+1)\times D} \tag{1}$$

$$z_l^{'} = MSA(LN(z_{l-1})) + z_{l-1}, \quad l = 1, 2, \cdots, L \tag{2}$$

$$z_l = MLP(LN(z_l^{'})) + z_l^{'}, \quad l = 1, 2, \cdots, L \tag{3}$$

$$y = LN(z_L^0) \tag{4}$$

One of the key blocks of Transformer is MSA, which comprises h parallel self-attention (SA) blocks, enabling the model to jointly attend to information from different representation subspaces at different positions. Each MSA block accepts an input sequence $z \in R^{(N+1)\times D}$, which is linearly projected to q, k and v by multiplying a weight matrix U_{qkv}. The attention weight A is then computed as the softmax of the dot product of q and k^T, divided by $\sqrt{D_h}$. The SA is a weighted sum of over all values v in the sequence, where A serves as the weight matrix.

$$[q, k, v] = zU_{qkv}, \quad U_{qkv} \in R^{D\times 3D_h} \tag{5}$$

$$A = softmax(qk^T/\sqrt{D_h}) \tag{6}$$

$$SA(z) = Av \tag{7}$$

The output of MSA is the concatenation of h SA blocks multiplied by a projection matrix U_{msa}.

$$\begin{aligned} MSA(z) =& [SA_1(z); SA_2(z); \cdots; SA_h(z)]U_{msa}, \\ & U_{msa} \in R^{h\cdot D_h \times D} \end{aligned} \tag{8}$$

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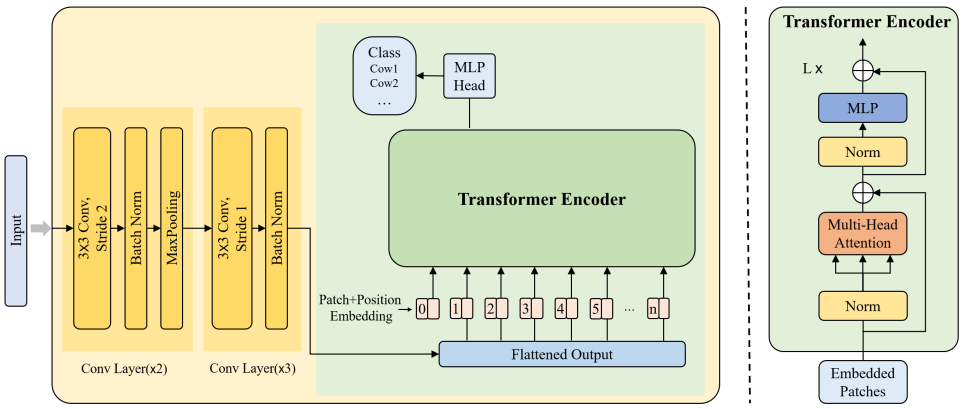

Fig. 1. Model overview. First, the cow face images are split into patches, and a position embedding is added to each patch. The sequence of patches along with the classification token a is then fed to a standard Transformer encoder. To enhance local field perception and have more inductive bias, we modified the patch generation method of ViT, inputting images of cow faces into a series of 3×3 convolution modules with stride 2 or 1 to extract embedded patches. Notably, after passing through the first convolution module, the number of image channels C = 3 increases to dim/16, and subsequently the number of channels doubles for each convolution module until C = dim.

B. Noise Identification

In recent years, large-scale annotated data are indispensable to the success of deep neural networks. However, when noise labels are mixed in the data, the networks believe all of the data are ground truth, and can easily overfit to noise, resulting in poor performance.

Unlike supervised learning (SL) assumes that all data are labeled correctly, SSL can take full advantage of the training data by utilizing samples and their labels that are likely to be correctly labeled and utilizing only samples which are potentially corrupted. In other words, the SSL in learning with noisy labels trains the network model using both clean and noisy samples, which are divided by SS on an epoch level or on a mini-batch level. The method leverages the clean labeled samples to guide the noisy samples, which are regarded as unlabeled samples. The process is illustrated in Fig.2.

In the SS stage, the whole noisy training set is divided into a labeled clean set $(\mathcal{X})$ and an unlabeled unclean (noisy) set $(\mathcal{U})$ on an epoch level or a mini-batch level. Specifically, the SS process involves calculating the loss of each sample and then dividing them into clean or unclean using a certain algorithm. One common approach is to regard samples with relatively small loss as clean, while those with high loss are regarded as unclean. Notably, different from other methods that only utilize the selected clean samples for training, we just discard the labels of unclean samples and exploit them in SSL.

During the SSL stage, clean samples with their annotated labels are treated as labeled data, while the remaining unclean samples whose labels have been discarded are regarded as noisy samples, treated as unlabeled data. SSL methods leverage the unlabeled data to improve model performance. One commonly used type of SSL is based on consistency regularization, which assumes that if a small perturbation is added to an unlabeled sample, the model predictions for that sample should remain consistent. Another popular SSL method is pseudo-labeling, which involves assigning predictions to unlabeled samples as their pseudo-labels and combining them with labeled samples to train the model. Besides these two methods, there are various other SSL algorithms and any one of the algorithms can be used to build SSL noise learning.

Based on the framework of SemiNLL, we have implemented an instantiation consisting of two independent networks that teach each other to avoid confirmation bias [19]. Specifically, during the SS stage, we simply select samples with relatively small losses as labeled data and discard labels of samples with larger losses as unlabeled data. The proportion of unlabeled data is determined by the noise rate. During the SSL stage, we specify the MixMatch as our SSL algorithm. In the MixMatch, the label for an unlabeled sample is guessed together by the two networks with a temperature sharpen applied to it. Then, multiple augmentations of the labeled samples $(\mathcal{X})$ and the unlabeled samples $(\mathcal{U})$ are concatenated and shuffled. Then, the original samples are interpolated with the shuffled samples. For two samples with their labels (x_1, p_1) and (x_2, p_2), the mixed sample is (x', p') and is computed by:

$$\begin{aligned} \lambda &\sim Beta(\alpha, \alpha) \\ \lambda' &= max(\lambda, 1-\lambda) \\ x' &= \lambda' x_1 + (1-\lambda') x_2 \\ p' &= \lambda' p_1 + (1-\lambda') p_2 \end{aligned} \tag{9}$$

After mixed, the $\mathcal{X}$ and $\mathcal{U}$ are transformed into $\mathcal{X}'$ and $\mathcal{U}'$. The loss function for $\mathcal{X}'$ is the cross-entropy loss, while the loss function for $\mathcal{U}'$ is the mean squared error with a factor λ_u to control the strength of the unsupervised loss:

$$\begin{aligned} \mathcal{L}_{\mathcal{X}} &= -\frac{1}{|\mathcal{X}'|} \sum_{x,p \in \mathcal{X}'} \sum_{c} p_c \log\left(\mathrm{p}^{\mathrm{c}}_{\mathrm{model}}(x;\theta)\right) \\ \mathcal{L}_{\mathcal{U}} &= \frac{1}{|\mathcal{U}'|} \sum_{x,p \in \mathcal{U}'} \|p - \mathrm{p}_{\mathrm{model}}(x;\theta)\|_2^2 \end{aligned} \tag{10}$$

where p_c is the one-hot label value for class c, $\mathrm{p}^{\mathrm{c}}_{\mathrm{model}}$ is the model output softmax probability for class c, p is the guessed label.

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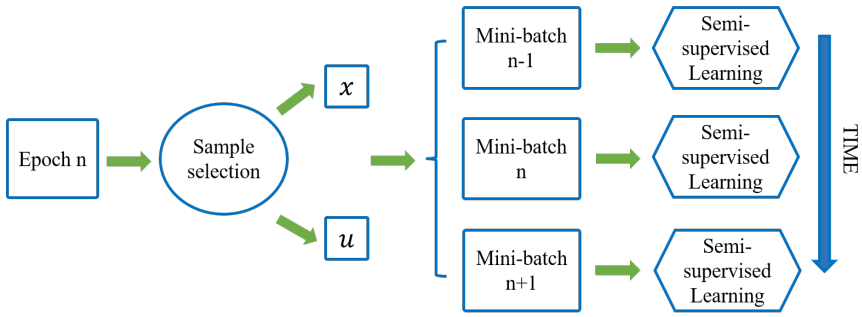
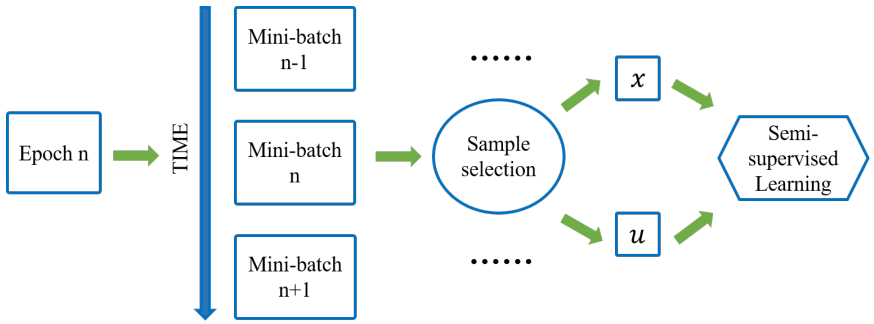


(a) SemiNLL with the SS on an epoch level



(b) SemiNLL with the SS on a mini-batch level

Fig. 2. Two SemiNLL frameworks

To avoid that the network tend to predict all samples to a single class under high noise rate, a regularization term (Tanaka et al., 2018; Arazo et al., 2019) [20] [21] is added to the loss function:

$$\mathcal{L}_{reg} = \sum_{c} \pi_c \log\left(\pi_c / \frac{1}{|\mathcal{X}'| + |\mathcal{U}'|} \sum_{x \in \mathcal{X}' + \mathcal{U}'} \mathrm{p}^{c}_{model}(x;\theta)\right) \tag{11}$$

where π_c is a uniform prior distribution.

The total loss function is:

$$\mathcal{L} = \mathcal{L}_{\mathcal{X}} + \lambda_{\mu}\mathcal{L}_{\mathcal{U}} + \mathcal{L}_{reg} \tag{12}$$

IV. Experiments

A. Datasets

We validate our proposed method on a private dataset consisting a total of 3556 images of cows. The dataset comprises images of 6 individual cows captured from 2 or 3 videos. To process the images, we manually annotated each cow image with a unique cow ID using the labelImg tool, which generate XML files according the face frame. Based to the XML files, we cropped out the cow face part and resized each image to 224×224 pixels. Examples of the cows can be seen in Fig.3. The resulting dataset was split into a training set containing 2982 cow face images and a testing set containing 574 cow face images, with the images in each set originating from different videos. To further evaluate the performance of our model, we set aside 30% of the testing set as a verification set, and used the remaining 70% as a new testing set.

B. Implementation Details

We utilized the ViT model as our classification network and made modifications to the way patch embeddings are generated, as described in Subsect.III-A. Considered that the cow face images of ours is a very small-scale dataset for ViT, we adjusted the hyper parameters of the model. Specifically, we set the input patch size to 32×32, the number of layers to 12, the hidden size to 192, the MLP size to 1024, and the number of heads to 3 with each head size set to 64. Additionally, all dropout rates were set to 0.1. Instead of using the 'cls' token, we opted to use the mean of tokens as our model classification token.

In our work, we experimented with only symmetric label noise generated by a symmetric noise transition matrix [22]. Each sample has a probability of p equal to the noise rate for transitioning to other classes. The probability of transitioning to any other class is $p/(C-1)$, where C denotes the number of classes. To train our network, we used AdamW with default parameters settings and cosine learning rate decay with T_{max} = 20, eta_{min} = 3e-4. The network was trained for 100 epochs and the initial learning was set to 6e-4 which was decreased to 3e-4 within 20 epochs followed by an increase to 6e-4 in the next 20 epochs. The two ViT models were trained from scratch without pre-training and had 5 warmup epochs for better convergence of the models. The Color jitter with a probability of 80% was used for training data augmentation. We trained our models with a noise rate of 0, 0.2, 0.4 and 0.6. In the presence of noise label, we set the λ_u to 25 and divided the training set on a min-batch level with a batch size

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(a) No.1 cow (b) No.2 cow (c) No.3 cow

(d) No.4 cow (e) No.5 cow (f) No.6 cow

Fig. 3. Examples of cow faces' images

of 64. When the noise rate was 0, we still considered 1% of the samples as unclean samples, with the λ_u set to 0, and divided the training set on an epoch level with a batch size of 16.

C. Performance Comparison

To evaluate the effectiveness of the proposed approach, we compare it with several methods under symmetric noise type. We save the model parameters that achieves the best accuracy on the validation set for testing . The results in the presence of noise are provided in Table I. For GCE loss function [23], we empirically set the parameter q to 0.7 and the k to 0.5. For APL loss function [16], we choose the NCE and NMAE with $\alpha = \beta = 1$. For Co-teaching [15], the forgetting rate is equal to the noise rate. Moreover, the experiment on Table II also show that ViT model can achieve a high test accuracy of 94.27% for cow face recognition.

TABLE I
COMPARISON WITH OTHER METHODS IN TEST ACCURACY (%) ON OUR COW FACE DATASET WITH SYMMETRIC NOISE RATE 0.2, 0.4 AND 0.6.

Method	Noise ratio		
	20%	40%	60%
Cross-Entropy	75.87	73.88	61.69
GCE [23]	84.82	79.85	67.91
APL [16]	86.82	75.12	71.89
Coteaching [15]	87.69	84.70	74.75
Ours	**92.79**	**92.04**	**91.79**

TABLE II
COMPARISON WITH OTHER METHODS IN TEST ACCURACY (%) ON OUR COW FACE DATASET WITHOUT NOISE.

Method	Test
Cross-Entropy	85.57
Ours	**94.27**

V. CONCLUSION

In the paper, a ViT-based noisy cow face recognition approach using the SemiNLL framework is proposed, with the aim of investigating the feasibility of ViT model in animal face recognition. The proposed method trains two ViT networks independently to teach each other and consists of two stages: SS and SSL. In the SS stage, noise training data are divided into labeled and unlabeled samples. In the SSL stage, both labeled and unlabeled samples are leveraged to train the networks. Moreover, we modify the way patch embeddings are generated, using a series of 3×3 convolution model to generate patches. Even though the network was train on a small cow face dataset, the proposed method also achieves a high test accuracy of 94.27% and robustness to noise on the task of cow face recognition.

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